Analysis of the Brazilian Electric Power System Considering the New 500 MW Interconnection with Uruguay

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***Abstract*— The new connection for the exchange of electrical power between Brazil and Uruguay, planned to come into operation in 2014, will have operational restrictions for import or export energy in the early years. The restrictions are due to unfavorable dispatch scenarios in the Brazilian Electric System. Until the entry into operation of the new coal-fired power plants, in the region of Rio Grande do Sul, there will be restrictions for the power exchange. In this work the conditions of service for this energy exchange are analyzed for several load scenarios. The simulations were performed using the Newton-Raphson method for the power flow solution.**

***Index Terms*— Power system interconnection, power flow analysis.**

I. Introduction

This new interconnection between Brazilian and Uruguayan electrical systems was authorized in 2010 [1], both for import and export of electric power. Initially the exchange of energy from Brazil to Uruguay (export) will come from a possible excess of available hydropower in the Brazilian hydroelectric system, or from not dispatched thermo-power plants. Therefore, only transmission lines are being constructed in Brazil for the operation of this interconnection.

Expected to start the operation in 2014, the new international interconnection will have a capacity of 500 MW in both directions. To make this energy interchange possible, it is essential to know the limitations for the sending and receiving of energy using this new interconnection, as well as the bottlenecks in the Brazilian and Uruguayan electrical systems. The exchange is planned for emergencies, surplus of power and lack of generation. In this paper only the limitations due to the Brazilian system will be analyzed, for the period between 2014 and 2017.

To attend the significant expansion of wind and thermal power generation in the state of Rio Grande do Sul (RS), various transmission system upgrades are planned to be implemented as of 2018. These upgrades will have great influence on the capacity of this interconnection between Brazil and Uruguay. As the schedule of these works still depends on the confirmation of the projected expansion, and there is uncertainty about the amount of energy planned to be sold in the coming auctions, this study did not analyze the interconnection beyond 2018. The details of the expansion of the transmission system carried out to reach these new plants and which will impact this connection as of 2018 can be seen in [2].

II. General Considerations

Cases of power flow obtained from the *Empresa de Pesquisas Energéticas* (EPE) were used as a base case. The files used for the simulations in *Anarede* [3] software were provided on 04/09/2013 [4].

The 525 kV bus at Melo was adopted as a PQ type with unity power factor. All the loads were represented as constant power [5], and thermoelectric units were dispatched as inflexible.

III. Brazilian Interconnection in South America

In South America, Brazil and Venezuela have a nominal frequency of 60 Hz, while in all the other countries the grid frequency is 50 Hz. In order to interconnect countries with different frequencies, it is necessary to use Frequency Convertor Stations (CF).

Brazil has electrical interconnection with Paraguay, Uruguay, Argentina and Venezuela, through the following interconnections [6]:

- Paraguay: Four 500 kV 50Hz Transmission Lines (TL) interconnecting the Paraguayan side of the Itaipu Hydroelectric Power Plant to the Substation Foz do Iguaçu in Brazil, which converts AC to DC and transmits to Ibiúna (Sao Paulo State), with a capacity of 6300MW;
- Uruguay: A 230 kV TL between Livramento (Brazil) and Rivera (Uruguay), with a capacity of 70 MW, and Frequency Converter Station in Rivera;
- Argentina: A Frequency Converter Station Uruguayana (Brazil), back-to-back, with a 132kV TL to Paso de los Libres (Argentina), with a capacity of 50 MW. Two 525 kV TL interconnections with Garabi (Argentina), where there is a Frequency Converter Station with a capacity of 2200 MW;
- Venezuela: A 230 kV TL between Boa Vista (Brazil) and Santa Elena (Venezuela), with a capacity of 200 MW. There is no need for Frequency Converter Station in this interconnection.

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IV. Interconnection Features

This new interconnection between the Brazilian and Uruguayan electrical systems will be capable of 500 MW of power transference from one system to the other, in both directions, because the Melo Frequency Converter Station is Back-to-Back type.

The connection will be made between the cities of Candiota (Brazil) and Melo (Uruguay), through a 525 kV transmission line, with 55 km on the Brazilian side between the cities of Aceguá and Candiota and 62km on the Uruguayan side between the city Aceguá and Melo [7]. This TL will be connected to one module of harmonic filters +65MVAr (60 Hz frequency) in the Candiota substation and one 92MVAr Reactor in the Melo substation.

In the Uruguayan side, the Melo Frequency Converter Station will be connected to the San Carlos substation through a 500 kV 50Hz transmission line. Fig. 1 shows in detail the region with the configuration for the year 2014, and Fig. 2 shows the simplified single-line diagram of the region. In Fig. 1, the red color is used to represent the 525 kV level, the green color represents 230 kV and the yellow color is used for 138 kV. Transmission lines that are under construction are shown with a dotted line.

The Melo Frequency Converter Station has 3 modules of harmonic filters (plus one spare one). Each module is +96 MVAr reactive power at frequency of 60 Hz. Fig. 2 shows these filters as shunts elements. SIN represents the connection to the main Brazilian transmission system and CF represents the Melo Frequency Converter Station connection to the Uruguayan grid.

The initial studies predicted the export of energy from Brazil to Uruguay, conditioned to the use of non-dispatched thermal power in the region or the excess of hydroelectric power that could not be used by the Brazilian loads [8]. Additional studies are currently being conducted by the Ministry of Mines and Energy (MME) for full integration of the Uruguayan and Brazilian electric markets.

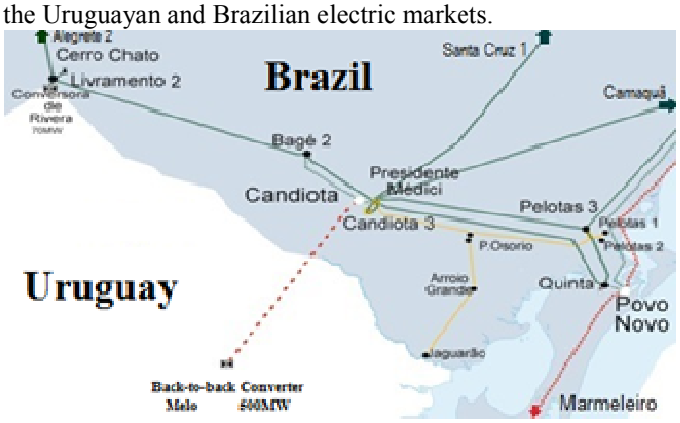


Fig. 1. Transmission lines and substations in 2014

V. Power System Upgrade

The entry into operation of the substation Candiota (CTA) is planned for 2014, with four -75 MVAr Reactors (one for reserve), three 224 MVA 525/√3 / 230/√3 / 13.8 kV single-phase autotransformers (ATF), and a 230 kV 10km Transmission Line to the substation of the Thermal Power Plant Presidente Medici (PME), with a capacity of 540 MVA.

In 2016 it is planned to start operations of the 230 kV transmission line between Bage2 and Candiota, with 290MVA normal capacity and 360MVA emergency capacity that should reduce undervoltage problems in southern Rio Grande do Sul during the contingency of 230 kV transmission line Bage2 / PME [9].

In 2018, various upgrades are planned in the region, depending on the projection of the expansion of wind and thermal generation units [2]. Two mineral coal fired plants, Seival and CTSUL, each with a capacity of 600MW, and a natural gas fired power plant near Rio Grande harbor with installed capacity of 1,200 MW are planned.

VI. Analysis of Integration in 2014

Between the years 2014 and 2017, this Brazilian-Uruguayan interconnection will operate with transmission restrictions depending on the dispatch of the Candiota III and PME power units. Fig. 2 shows the interconnection of plants Candiota III and PME to the Brazilian electric system, and the interconnection of these plants to SIN through the substation PME (9233 bus). The PME has two generation units: Unit A (9432 bus) and Unit B (9440 bus).

Fig. 2 shows the Single Line Diagram for the interconnection between Brazil and Uruguay, where only the main network elements of interest for this work are represented.

In any of the simulated cases the thermal plant of Uruguaiana is in operation, but its entry into operation reinforces the 230 kV system in the western border of Rio Grande do Sul, improving performance for contingencies and voltage control, using the excitement of the plant machines. With this, its entry into operation contributes to a better performance of energy exchange between Brazil and Uruguay via the Melo Converter Station.

The Candiota III generates 315 MW with the Medium load cases. The PME generates 25 MW (Unit A) and 90 MW (Unit B) for light load cases. For the Medium load cases PME generates 81 MW (Unit A) and 302 MW (Unit B). And during heavy load cases when Rio Grande do Sul is importing power from the SIN, the PME generates 25 MW and 90 MW, whereas when Rio Grande do Sul is exporting power to the SIN it generates 81 MW and 302 MW.

VII. Cases of Northern Brazil Exporter and Northern Brazil Importer

Depending on the hydrological cycle of the plants and the electrical system configuration, power exchanges between Brazilian subsystems change. Mainly there are two cases: a case where the north exports power to the SIN (Nexp) and a case where the north imports from the SIN (Nimp) [2].

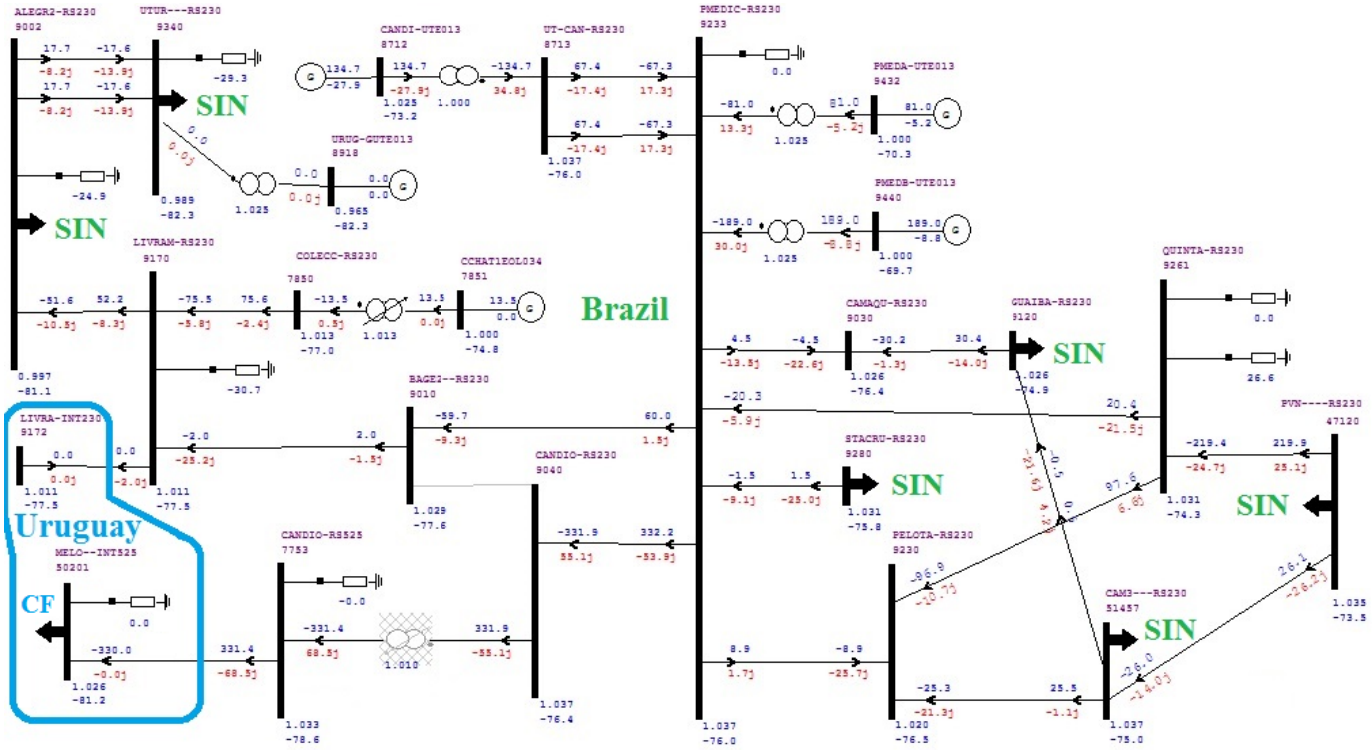


Fig. 2. Single-line diagram of the transmission system close to the interconnection.

Cases with Nexp refer to the humid period in the north, with high hydrological activity in the northern basins of Brazil. In this case the power comes from the north to the south through the SIN.

Cases with Nimp represent the dry period in the north, where there is power transmitted from the south to the north using the SIN.

Throughout the day the dispatch of the power plants of the SIN changes, and these two settings can happen on the same day, as shown in Fig. 3. Therefore it is necessary to examine this interconnection between Brazil and Uruguay considering both cases-Nexp and Nimp. In Fig. 3, the positive active power flow means that the south exports power to the north.

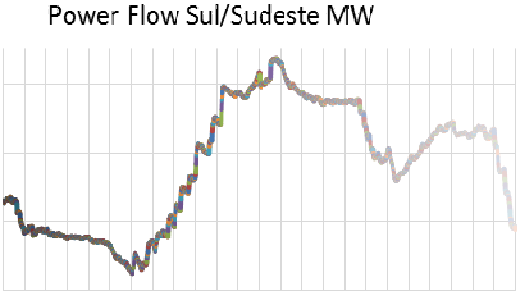


Fig. 3. Power flow from south to southeast system by 02/05/2013.

Analyzing the impact of these types of cases as they operate the interconnection, we can see in Table I that Rio Grande do Sul is importing power from the SIN (Nexp) in the EPE and ONS reference cases. Rio Grande do Sul receives more power in the Nexp than in the Nimp case due to the addition of Thermal and Hydro plants installed in the region.

Table I shows that there are differences in the exchange of energy from Rio Grande do Sul to the SIN, between the reference cases of EPE and ONS, which influence the dispatch of the plants and the conditions for the exchange of power between Brazil and Uruguay. While the EPE conducts long-term studies with 10-year horizons, ONS simulates medium-term horizons (4 years).

TABLE I. POWER FLOW FROM RIO GRANDE DO SUL TO SIN.

Base case study	EPE, year 2014					
	Setting North Exporter			Setting North Importer		
	Light Load [MW]	Medium Load [MW]	Heavy Load [MW]	Light Load [MW]	Medium Load [MW]	Heavy Load [MW]
EPE PDE 2021	-1.794,8	-4.091,6	-3.805,8	-831,4	-2.941,0	-2.172,9
EPE PDE 2022	-2.638,7	-3.706,3	-3.826,3	-761,5	-2.727,1	-2.103,7

ONS, setting PAR 14-16, year 2014				
Light Load [MW]	Medium Load [MW]		Heavy Load [MW]	
June-2015	June-2015	October15 - March16	April - September15	October15 - March16
-883,7	-2.454,6	-5.014,7	-2.622,3	-4.521,6

In studies of EPE, wind generation is dispatched considering its capacity factor, while the other plants are dispatched analyzing hydrological data. ONS studies consider

the typical wind regimes for the dispatch of the wind farms. These conceptual differences in generation dispatch are reflected in the difference of the exchange seen in Table I.

VIII. Analysis of Contingencies

Contingencies in the interconnecting neighbor buses were simulated. The interconnection bus in the Brazilian grid is 525 kV bus Candiota (CTA). The simulated contingencies that affected the capacity of power transmission across the interconnection are listed in Table III.

During the analysis of contingencies the electrical performance should be checked after disconnecting an element of the system. It is common practice to allow the voltage regulation of PV buses that are controlled by generation units, synchronous and static compensators, operation of the automatic tap changers of transformers or operational measures that depend on human actions [5].

After the use of all possible operative measures for voltage regulation, the load limits of transformers and transmission lines must be checked. When maximum emergency loading is violated or the maximum voltage variation shown in Table II is achieved it is time to plan the system upgrade.

To regulate the voltage in the Candiota region, the most frequently used measures were:

- Connection or disconnection of reactors and capacitors;
- Control of excitation of synchronous generators of hydroelectric and thermoelectric power plants in the region;
- Control the excitation of wind turbines with wound-rotor, synchronous multi-poles generators in the region of the Santana do Livramento;
- Changing the taps of the 525 kV/230 kV transformers.

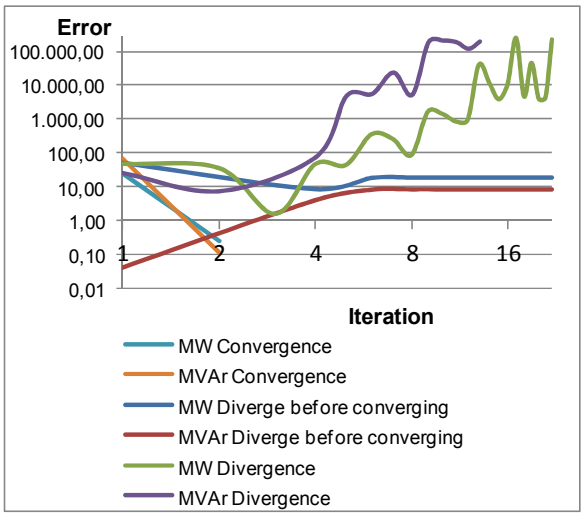


Fig. 4. Typical error found in the simulations.

Table II. Nominal line voltages at 60 Hz.

Nominal operating voltage	Normal operating condition		Operative emergency condition	
(kV)	(kV)	(pu)	(kV)	(pu)
< 230	–	0,95 a 1,05	–	0,90 a 1,05
230	218 a 242	0,95 a 1,05	207 a 242	0,90 a 1,05
345	328 a 362	0,95 a 1,05	311 a 362	0,90 a 1,05
440	418 a 460	0,95 a 1,046	396 a 460	0,90 a 1,046
500	500 a 550	1,00 a 1,10	475 a 550	0,95 a 1,10
525	500 a 550	0,95 a 1,05	475 a 550	0,90 a 1,05
765	690 a 800	0,90 a 1,046	690 a 800	0,90 a 1,046

Source: ONS module 23.3, page 11 [5].

Table III. Contingencies analyzed.

TL 525 kV	CTA/NSR; NSR/PVN; MML/SVP; PVN/MML;
TL 230 kV	ALE2/LIV2; ALE2/MBR; ALE2/SVI; ALE2/URU5; ALE2/UTUR; BAG2/LIV2; CAM/GUA2; CAM3/PEL3; CAM3/PVN; CTA/PME; CCH/LIV2; CIN/CHA; CHA/SCH; GUA2/CAM3; GUA2/CIN; GUA2/ELD; PEL3/PME; PEL3/QUI; PME/BAG2; PME/CAM; PME/QUI; PME/SCR1; PVN/QUI; SCR1/CHA; UDFR/UITA; UITA/CDL; UITA/PPE; UITA/SCR1; UITA/UPRE.
TF 525/230 kV	SE's CTA, PVN.
TF 230/138 kV	SE's PEL3, PME, QUI.
Reactor	- 150 MVAr na SE PVN 525 kV; - 75 MVAr na SE CTA 525 kV; - 30 MVAr 230 kV: PME, LIV2; - 25 MVAr 230 kV: QUI, ALE2.
Capacitor	25 MVAr na SE QUI 230 kV.
Generation rejection	Candiota III, 1 generator PME B, Cerro Chato Wind farm Santa Vitória do Palmar e Chuí Wind farms.

a Generations means Candiota III and PME A and B.

Table IV. Power Exchange restrictions MW.

Case	Restriction	Generation[a]
Light load, Nexp	No restrictions	115 MW
Light load, Nimp		
Medium load, Nexp	No active power from Brazil to Uruguay	Min 735 MW
	Max active power from Uruguay to Brazil	698 MW
Medium load, Nimp	No restrictions	Max 618 MW
	Max 420MW	698 MW
	Max power from Brazil to Uruguay 460MW	Min 735 MW
Heavy load, Nexp	Max power from Brazil to Uruguay 330MW	Min 150 MW
	No restrictions	Min 220 MW
	No restrictions when power flows from Uruguay to Brazil	150 MW
Heavy load, Nimp	Max power from Uruguay to Brazil 330MW	Max 698 MW
	No restrictions when power flows from Uruguay to Brazil	
	No restrictions	Max 638 MW

IX. Problems of Convergence

The *Anarede* software calculates the power flow of AC and DC systems with successive iterations until the error of the active and reactive power is lower than a given tolerance [3].

Initially, the operating state of the DC system is determined. The CC system convergence usually happens in the first iteration, then the algorithm searches for the convergence of the AC system. To solve the AC system issue the default method is the Fast Decoupled Newton-Raphson method.

The worst operational cases for the Melo converter Station (Medium and Heavy load Nexp) gave angles of the buses in the region between -60° and -90° (compared to the reference bus at Ilha Solteira). The closer the angles to -90° the higher incidence of problems of convergence in iterative calculation of the AC power flow.

Fig. 4 shows graphically the most common forms of the error variation during the iterations for the solution of the AC power flow using the method of Newton-Raphson. Before including Melo when the angles are close to 0° the algorithm starts with small errors and converges fast. This situation is shown in Fig. 4 (light blue and orange lines).

In some cases, allowing for greater tolerance by changing the error range can solve the problem of convergence of the method (shown in Fig. 4 by the blue and brown lines). Note that in this case good sense should be used to define the values for error tolerance of active power in MW and reactive power in MVAr, because in this case the solution is not the point of absolute minimum error, but a local minimum. Due to the formulation of the Newton's method some small changes can lead to the divergence of the solution, and define that the system is operating in a numerically unstable region.

Depending on the tolerance value, convergence of the method can occur for divergent systems, as can be seen by the green and purple lines in Fig. 4. In these cases, where the error will naturally diverge if a standard 0.1 tolerance is set, a convergence is achieved when the tolerance is set to 10.

A simple test to check if the convergence of the power flow is not about to diverge is to make a small change in a neighboring bus, and then undo this change. The system must return to the same pre-change voltage values. An example of this method is to switch the –30 MVAr reactor at PME.

In all simulations changes that lead to few iterations of Newton-Raphson, indicating that convergence occurred for the minimum global error of the numerical system were searched for. Hence, some measures were unavoidable, such as: the connection of the 525 kV bus of the Melo Converter Station with the appropriate initial conditions (the same voltage as 525 kV Candiota bus), and the switching off of the automatic tap change of the 525/230 kV transformer at Candiota (using the option *CTAP* in software *Anarede*).

X. Restrictions to Power Transference

In the simulations, the worst cases are observed for sending 500 MW from Brazil to Uruguay in the Medium and Heavy loads with Nexp as shown in Table IV.

The difference between the medium and heavy load cases is mainly due to the power factor, which on medium load is more inductive than the heavy load case in Rio Grande do Sul.

In Table IV, there is only one case where it is not possible to send energy from Brazil to Uruguay, medium load with Nexp. In this case there are restrictions in the region, and even maximizing the generation of Presidente Medici and Candiota III there is still a need for load shedding in the region to withstand any generation rejection above 75MW.

XI. Results Summary

This power exchange was analyzed for the main 6 cases of the Brazilian electrical system: (Light, Medium or Heavy load) and power flow between subsystems south and southeast (Nexp or Nimp).

Several contingencies were simulated according to the N-1 criterion (ONS [5]). Transmission lines, transformers, reactors and generation rejection were analyzed. In Table III the contingencies simulated are listed, and a resume of the power exchange restrictions is found in Table IV.

The most severe contingencies that caused more restrictions were:

- Rejection of Candiota 3 plant when it is generating above 300MW;
- Loss of 230 kV transmission line PEL3 / PME, causing overloading of 230 kV transmission line PME / QUI;
- Loss of 230 kV transmission line PME / QUI, causing overload of 230 kV transmission line PEL3 / PME.

XII. Conclusion

The limitation of power exchange between Brazil and Uruguay for Medium and Heavy load cases requires additional studies, due to the fact that, depending on the dispatch and schedule changes in real time, there may be reverse power flow between the Rio Grande do Sul and the SIN.

As an example of restriction to export 500 MW to Uruguay during the Medium and Heavy load cases, if the interconnection is sending power to Uruguay in a Medium Load case with Nimp and there is a reversion to Nexp, the power sent to Uruguay will be reduced to zero.

In order to improve the operation of this interconnection, the installation of reactive shunt elements in the region is needed and must be scheduled for future system upgrades.

When Uruguay exports 500 MW to Brazil, the Brazilian system performs better in all the simulated cases compared with the situation in which Brazil exports power to Uruguay.

In all cases where Brazil sends power to Uruguay, the addition of generation in the Brazilian region showed an improvement in the exchange performance during the simulated contingencies.

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